

Decision Intelligence for Defense Logistics

AI, DIGITAL TWIN, AND LARGE-SCALE DIGITAL TRANSFORMATION
FOR MISSION-READY LOGISTICS



AI-DRIVEN DECISIONS

Smarter planning. Faster response.



DIGITAL TWIN VISIBILITY

Real-time insights across the enterprise.



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End-to-end logistics. Global impact.



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Glossary

Glossary of Key Terms

EDM (Entity Digital Model) Vector Intelligence- An approach that rebuilds each item, supplier, depot, and route as a time-aware digital twin and represents its state as a high-dimensional vector, so it can be compared over time and against peers.

BEI (Behavioral Entity Intelligence)- The behavioral-intelligence approach, applied across demand, supplier-risk, and network early-warning, that analyzes entities, their relationships, and the network over time to surface slow, directional change.

NSN (National Stock Number)- The standard identifier for a supply item; the first four digits are the Federal Supply Class.

DODAAC- Department of Defense Activity Address Code; identifies the activity or location associated with a requisition.

P10 / P50 / P90- Points on the forecast range: the P90 is the level demand is not expected to exceed about 90 percent of the time, used here as the planning (buy) quantile; the P50 is the midpoint.

Calibration (ECE, Brier score)- Measures of whether predicted probabilities match real frequencies; lower values mean the forecast ranges can be trusted.

Hurdle model- A two-stage forecast that first predicts whether demand will occur at all, then how much, which suits intermittent demand.

Drift / CUSUM- Drift is a gradual, directional change in behavior; CUSUM is a change-detection method that accumulates small shifts until they become significant.

Cohort- A peer group of similar items or suppliers; behavior is judged relative to the cohort, and new items borrow from their nearest behavioral twins.

Survival model- A method that estimates the time until an event, such as a supplier disruption, and how that risk develops over 30, 60, and 90 days.

Embedding- A numerical vector that captures the meaning of serialized entity state; here, it generates most of the engineered features (zone, drift, and cohort features).

Criticality code- An item designation (for example, C or B) reflecting mission importance, used to set how protective the planning level should be.

SHAP- A method that explains a model's output by attributing it to individual input features, so each forecast or risk score is traceable.

MASE (mean absolute scaled error)- A scale-free forecast-accuracy measure suited to intermittent demand; below 1 means better than a naive baseline.

Pinball loss- The standard scoring rule for quantile (range) forecasts, such as the P90; lower is better.

SBOM (software bill of materials)- A structured list of the components in a software product, used to track supply-chain and component risk.

1 Executive Summary

The Challenge: Fragmented Data and the Loss of Decision Time: Today, critical operational intelligence lives in silos. Information about a part, supplier, depot, route, or user is scattered across disparate systems, tables, and structures, each holding only a fragment. Because legacy mathematical models confine analysis to individual attributes, one field at a time, decisions are consistently made on isolated slices of data rather than the complete, comprehensive reality of the entity. We rarely see the full picture.

For the DLA, that gap has a mission cost. It operates in a contested logistics environment where adversarial exploitation and disruption can emerge through demand volatility, supplier fragility, sub-tier dependency, counterfeit exposure, sourcing shifts, route vulnerability, cyber-enabled disruption, or correlated movement across the defense industrial base. In this environment, the mission risk is not simply a late shipment, an inaccurate forecast, or a delayed supplier alert. The mission risk is the loss of decision time, the interval between the first observable behavioral signal and the point at which readiness, mission assurance, or operational resiliency is affected. Current planning, forecasting, and supplier-monitoring methods remain necessary, but they can leave DLA reacting after thresholds are breached rather than anticipating where demand, supplier behavior, inventory posture, or network exposure is moving. The requirement is a stronger decision-intelligence layer that preserves decision time and supports decision advantage before operational consequences are fully visible.

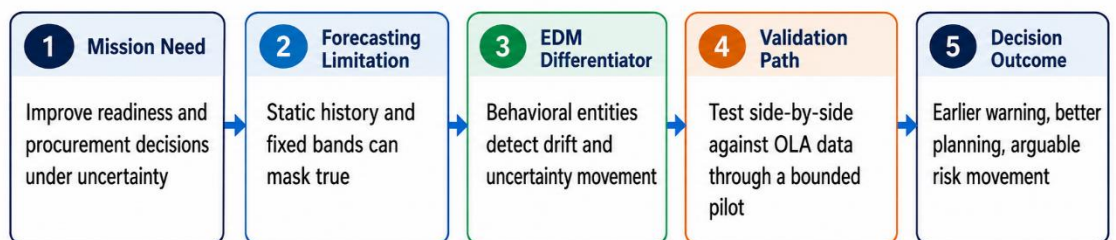
The Innovation: Bringing Self-Attention to the Decision Domain: Our innovation, Entity Digital Model (EDM) Vector Intelligence and Behavioral Entity Intelligence (BEI), collectively EDM / BEI (Patent Pending), solves this fragmentation by

representing every entity and its network of connections as a single, holistic behavioral digital twin.

The breakthrough is taking the core mathematical engine behind Large Language Models, the Transformer architecture's self-attention mechanism, out of the text domain and applying it to

operational decision-making. Just as self-attention weighs the words in a sentence against one another to derive context, EDM / BEI weighs every element of an entity's operational data against every other element. Defining the entity as a digital twin is what captures that mechanism. Each item-location, supplier, depot, route, relationship, and network becomes a rich, high-dimensional behavioral representation within a shared digital space; its movement is measured over time, compared against its own history and peer cohorts, aligned to known supply-chain concepts, and translated into explainable planning or risk signals. Figure 1 illustrates this shift from backward-looking metrics to forward-looking, mission-relevant decision intelligence.

From Metrics to Mission Decision Intelligence



Executive takeaway

Use existing data to move from backward-looking metrics to earlier, more decision-useful signals.

Figure 1: Metrics to Mission Decision Intelligence

What Makes EDM / BEI Unique?

- **Unified entity space:** every entity type, from parts and suppliers to depots and users, is represented in one shared space, so cross-entity relationships become measurable rather than implied.
- **Direction over threshold:** it measures where behavior is heading, surfacing slow, sub-threshold changes that magnitude-based methods miss by design.
- **One foundation, many domains:** the same representation and behavioral signals carry across very different mission domains, making this a reusable platform rather than a single-purpose tool.

Proof of Concept: Generalizable Impact Across Domains: To validate the versatility of this foundation, we tested EDM / BEI across three distinct mission domains using synthetic data. The same behavioral signals were carried across all three, which is strong evidence that this is a generalized capability, not a narrowly tuned point solution. Early results are highly encouraging:

For the mission, this means tighter, better-justified procurement, earlier warning that protects readiness, and faster, explainable decisions, all delivered from one reusable foundation rather than a new tool built for each problem. The detailed evidence is presented in Section 2, “Proof of Concept: Decision-Value Evidence.”



Our Capabilities

- ✓ **Demand Forecasting:** EDM Vector Intelligence reduced the eight-week procurement buy by 40 to 54 percent on worked items while fully covering realized demand (for example, right-sizing a 379-unit adapter plate buy to 225 units, preventing roughly \$460,000 in over-procurement). It also produced forecasts for items that legacy methods report as zero, catching demand they miss entirely.
- ✓ **Supplier Risk:** Flags higher-risk suppliers and estimates time-to-failure, with explainable behavioral drivers.
- ✓ **Logistics Network Early Warning:** BEI surfaces slow, sub-threshold, directional change across the logistics network, including supplier behavior, sourcing shifts, route exposure, and correlated movement across the industrial base, giving warning before disruption reaches a readiness threshold. The underlying detection principle was most rigorously validated in a cybersecurity proof of concept, presented here purely as an analogy to demonstrate that it works.

The Recommended Next Step and Path Forward:

The recommended next step is a bounded pilot that runs current DLA baselines and EDM / BEI outputs side by side across selected NSNs, suppliers, depots, and mission-relevant scenarios. Use Case 1, Demand Forecasting and Inventory Planning, is the prioritized starting case because it provides the clearest path to validate forecast quality, uncertainty management, procurement-plan accuracy, early warning, and planner usability against existing DLA data. From there, the same operating model extends to supplier-risk and logistics-network early-warning use cases, in support of broader mission assurance, operational resiliency, supply-chain risk mitigation, and decision advantage.

Critically, the pilot builds the data foundation and the behavioral digital-twin representation once. On that same foundation, the platform is designed to address a mapped roadmap of 17 identified use cases across four horizons (demand, supply and lead-time risk, network and resilience, and portfolio intelligence), each reusing the data, embeddings, and features already built, so every wave is faster than the one before.

Beyond the demand pilot, these include cold-start forecasting, stockout prediction, lead-time estimation, supplier-capability discovery, route-bottleneck identification, cascading-failure and combined-shock analysis, drift early-warning alerts, item-health watchlists, and semantic substitute search. The path forward is to map the attributes that matter for each new scenario, simulate the corresponding digital twins, and demonstrate how wide the foundation scales.

2 Proof of Concept: Decision-Value Evidence

The current proof of concept demonstrates that EDM/BEI can produce decision-relevant outputs across DLA-aligned demand forecasting, supplier risk, and network early-warning scenarios. The results should be read as proof of concept, not as operational proof on DLA data. They show that the model architecture can generate calibrated planning ranges, supplier-risk discrimination, relationship-level visibility, explainable drivers, and early warning signals, warranting validation through a bounded pilot using DLA operational data.

EDM turns forecasting into planning intelligence. It models each item-location as a behavioral entity and uses demand, supply, inventory, volatility, and context signals to adjust the P10 to P90 planning interval. The initial assessment read 16 to 23 monthly snapshots, measured behavioral drift, aligned that drift to supply-chain concepts, and produced calibrated planning ranges for procurement and early-warning decisions. Because this proof-of-concept window spans roughly two years, it covers only about two annual cycles; full seasonal and exercise-cycle validation require the longer DLA demand history and is a pilot data-readiness item.



The core innovation is not another forecast but a new way of representing the problem. Each item and depot are rebuilt as digital twins using EDM Vector Intelligence, with their states embedded in high-dimensional vector spaces. Raw requisition, supply, and inventory records become a living, behavioral entity rather than a flat table. That representation is what makes advanced operations possible, including behavioral drift detection, concept alignment, peer-cohort transfer, and full distributional simulation, techniques that generic tabular data cannot support and that let the approach answer questions a conventional forecaster cannot.

Of the three representative items, EDM Vector Intelligence substantially reduced over-procurement against the moving-average baseline. For an adapter plate, it cut the 8-week P90 plan buy from 379 to 225 units, roughly 40 percent lower and about \$460,000 less over-procurement; for a needle bearing it cut the buy from 1,184 to 545 units, roughly 54 percent lower and about \$48,000 less. It also corrected the opposite failure: for a valve train kit with no recent history, the baseline forecast was zero while EDM inferred 95 units from similar items, catching demand that legacy methods miss entirely. These are per-item results on synthetic data and would be re-validated on DLA data.

EDM Turns Forecasting into Planning Intelligence

EDM models each item-location as a behavioral entity and uses demand, supply, inventory, volatility, and context signals to adjust the P10 to P90 planning interval.

- Read 16 to 23 monthly snapshots and measured behavioral drift
- Aligned drift to supply-chain concepts and produces calibrated planning ranges for procurement and early warning
- Adapter plate: 8-week P90 buy cut from 379 to 225 units, about \$460,000 less over-procurement, while still covering realized demand
- Needle bearing: 1,184 to 545 units, about \$48,000 less; valve train kit: 0 to 95 units, catching demand legacy reports as zero.

Those distributions are trustworthy because the model's probabilities are tightly calibrated. On the occurrence stage, which predicts whether demand will appear at all, it reaches an Expected Calibration Error (ECE, the gap between predicted probabilities and observed frequencies, where lower is better) of 0.0322 and a Brier score (the mean squared error of a probabilistic prediction, where lower is better and zero is perfect) of 0.0885. Calibration at this level is what lets a planner act on the full range rather than a single estimate, and it is the foundation of the procurement and early-warning decisions the twin is built to support. These results were established on synthetic data and are ready for validation on DLA operational data.

BEI converts supplier risk into behavioral intelligence. It models supplier risk across connected behavioral entities, including suppliers, NSNs, locations, routes, manufacturers, countries of origin, contracts, and sub-tier

BEI Converts Supplier Risk into Behavioral Intelligence

BEI models supplier risk across connected behavioral entities, including suppliers, NSNs, locations, routes, manufacturers, countries of origin, contracts, and sub-tier relationships.

- Surfaces interpretable drivers: quality incidents, recent order volume, on-time rate, worst delay, and critical-NSN count
- Estimates how risk develops over 30, 60, and 90-day horizons, alongside a point-in-time view
- Monotonic constraints keep the logic interpretable: stronger on-time performance lowers risk; quality incidents, sole-source exposure, and severe delays raise it
- Presented as a qualitative proof of concept requiring real-data validation

relationships, and surfaces interpretable drivers such as quality incidents, recent order volume, on-time rate, worst delay, and critical-NSN count. Modeling these as connected entities lets the approach see how supplier behavior and relationships are moving, rather than only whether a current-period metric has crossed a threshold.

The supplier-risk approach moves beyond static supplier scoring by evaluating how risk develops through supply-chain behavior and relationship patterns. Rather than treating risk as a single supplier attribute, BEI analyzes how operational performance, sourcing exposure, delays, and dependency patterns combine to affect mission readiness. Alongside a point-in-time risk view, the survival model estimates how risk develops over 30,

60, and 90-day horizons, so action can be taken before the window closes. The logic is built to be interpretable: monotonic constraints ensure that stronger on-time performance lowers estimated risk while quality incidents, sole-source exposure, and severe delays raise it, and explainable drivers show why a supplier is flagged. Applied as a full behavioral model, the same representation extends to relationship and sub-tier exposure and to drift that surfaces before current-period metrics change, which is where the approach adds the most for supplier risk.

BEI Detects Mission Risk Before Thresholds Are Breached

BEI provides a cross-domain early-warning framework that analyzes entities, relationships, and networks over time.

- Describes each entity along nine behavioral dimensions and applies three analytical layers (entity, relationship, network)
- In a synthetic cybersecurity proof of concept, we identified all four embedded long-duration attack campaigns at an 8.1% false-positive rate, where conventional anomaly methods caught at most one
- Offered as a transferable detection principle, that slow, persistent, sub-threshold signals are discoverable, and a hypothesis to validate on DLA data; the 8.1% operating point was ground-truth-selected, a reporting benchmark, not a deployable threshold
- Relevant to DLA because supply-chain and readiness risks may stay within nominal thresholds while their behavioral direction changes

This early-warning approach surfaces weak signals that conventional monitoring may miss when activity still appears normal on the surface. The proof of concept used synthetic telemetry covering 250 users, 485 days, and approximately 14 million events, providing a large test environment for evaluating long-duration behavioral change. The benchmark showed stronger campaign detection than conventional anomaly methods: the z-score method identified one campaign, while the local outlier factor, Isolation Forest, and the one-class support vector machine identified none. For DLA, the takeaway is that risk may develop gradually before it triggers a standard rule, metric, or threshold, and BEI helps identify changing patterns early enough to support investigation, prioritization, and readiness-focused response.

3 The DLA Decision Problem

As the Nation's Logistics Combat Support Agency, DLA must sustain warfighter readiness across a global supply chain that is increasingly nonlinear, relationship-dependent, and exposed to disruption. In a contested logistics environment, adapting faster, resolving supply-chain friction earlier, and preserving decision time are essential to mission assurance, operational resiliency, and U.S. military superiority. Current planning, forecasting, and supplier-monitoring methods remain necessary, but they can leave DLA reacting after thresholds break rather than seeing where demand, supplier behavior, inventory posture, or network exposure is moving. The requirement is a stronger decision-intelligence layer that improves supply-chain risk mitigation and supports decision advantage before operational consequences are fully visible. Forecasting, supplier monitoring, inventory planning, and readiness support are just a few of its many key activities. In conventional analytics, these are usually approached as separate problems, each with its own model and data, even though they are operationally coupled: a demand signal may rise because of mission activity, while the associated risk is governed by supplier capacity, depot position, lead time, part criticality, and network dependency. A supplier may appear compliant in aggregate while a single supplier-part-country relationship deteriorates, and a network may appear healthy in individual metrics while several nominally independent suppliers share a common fragile sub-tier source.

The recurring limitation is timing. Conventional methods tend to detect a consequence after the underlying behavior has already changed: a moving average responds once demand history shifts, a supplier score responds once delivery, quality, or financial thresholds deteriorate, and a network review responds once dependencies become visible through disruption. These methods remain necessary but are not sufficient in mission environments, where latent change can translate into procurement, readiness, or mission risk before a threshold-based alert is triggered.

The problem is therefore broader than model accuracy. DLA requires earlier and explainable signals that inform specific questions: what should be stocked, where risk should be buffered, which suppliers are drifting, which

relationships are changing, which latent dependencies are material, and what should be validated before a larger operational commitment is made. The appropriate analytical target is not an additional dashboard indicator but a layer that characterizes where demand, supplier behavior, inventory exposure, and network risk are moving.

These limitations do not indicate that current methods lack value; they indicate where current methods should be complemented. The proposition advanced here is that DLA can retain transparent baselines while adding a behavioral layer that measures drift, characterizes its direction, and supports controlled validation before procurement or readiness consequences become observable.



Why Traditional Analytics Fall Short?

- ✓ **Threshold insensitivity:** Rules and supplier scores can stay green while sourcing, relationship, or network risk changes underneath them. (e.g., a supplier keeps meeting its on-time target, so its scorecard stays green, even as its lead times creep up and it shifts to a higher-risk country of origin)
- ✓ **Magnitude-only measurement:** Conventional anomaly models measure how much a metric change, not what direction the entity is becoming. (e.g., a supplier's total order volume holds steady while its mix shifts toward critical, sole-source parts; the quantity is unchanged, but the risk direction has moved)
- ✓ **Single-entity scope:** A supplier, NSN, depot, or user may look normal individually, while the relationship between entities has changed. (e.g., two suppliers each look healthy on their own, yet both depend on the same sub-tier manufacturer, so one disruption takes out both)
- ✓ **Static uncertainty:** Static planning bands cannot narrow when demand calms or widen when volatility and disruption risk increase. (e.g., a fixed band holds a wide buffer through months of calm demand, then fails to widen fast enough when a mission surge hits)
- ✓ **Delayed response:** Backward-looking methods react after the data changes, leaving less time to buffer inventory, investigate supplier drift, or reduce mission exposure. (e.g., a forecast reacts only after consumption has already spiked, leaving little time to position inventory or investigate the supplier)

4 Technical Foundation - Behavioral Entities, Drift, and Decision Signals

Behavioral Entity Intelligence, as shown in Figure 2, represents an observable subject as a profile that evolves over time rather than as a static record. An entity may be an item-location, supplier, NSN, depot, country-of-

origin relationship, user, device, route, manufacturer, or network cluster. The representation captures the entity's attributes, its activity, its change over time, its spatial setting, its relationships with other entities, its sequential and periodic behavior, and the context that gives the behavior meaning.

The central premise is that consequential risk frequently manifests as a

One BEI / EDM Architecture Across Three DLA Use Cases

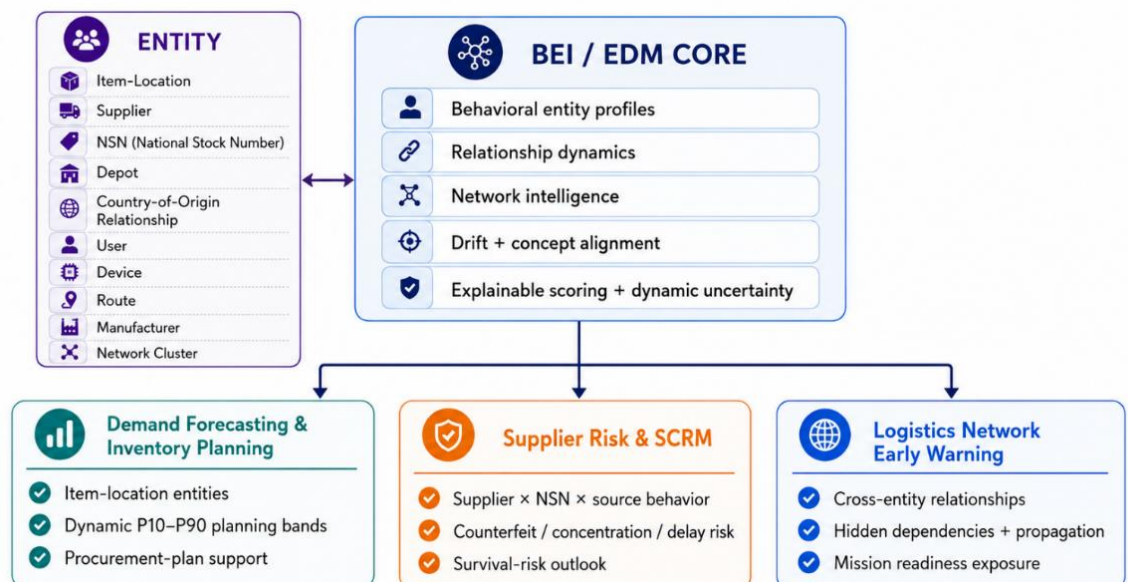


Figure 2: One BEI / EDM Architecture Across Three DLA Use Cases

change in what an entity does rather than in how much it does. A supplier may deliver a constant volume while shifting its sourcing toward a higher-risk geography; an item may exhibit sparse historical demand while its behavioral trajectory moves toward a disruption; and a network may present normal individual nodes while several relationships drift in a common direction. Conventional metrics quantify magnitude; BEI additionally characterizes the direction of behavioral change

4.1 Key Behavioral Dimensions

In the current proof of concept, each entity is described along nine key behavioral dimensions: identity, behavior, time, space, relationship, network, sequence, periodicity, and context. These are the dimensions that matter most for the use cases shown here, not a fixed or complete set. Which dimensions apply, and how many, depends on the business problem, the operational scope, and the context available in the data, and the framework adds more as those grow.

Dimension	What It Captures	DLA Relevance
Identity	Relatively stable attributes of the entity	NSN, supplier, Commercial and Government Entity (CAGE) / Unique Entity Identifier (UEI), depot, platform, criticality, commodity, contract, tier
Behavior	What the entity does across activity zones	Demand, delivery, quality, sourcing, inventory, financial, route, or access behavior
Time	Evolution, velocity, acceleration, and drift	Early warning when behavior changes before thresholds move
Space	Geography and topology	Depot position, country of origin, supplier location, theater, route, operational zone
Relationship	Pairwise interaction pattern	Supplier x NSN, Supplier x Country, NSN x Location, Supplier x Supplier
Network	Graph-level ecosystem intelligence	Hidden dependencies, common sub-tier exposure, correlated supplier movement
Sequence	Order of operations	Sourcing sequence, procurement event pattern, disruption sequence, maintenance path
Periodicity	Rhythm and cadence	Seasonal demand, exercise cycles, delivery cadence, and inspection pattern
Context	Environmental meaning	Mission posture, exercise activity, platform lifecycle, regulatory or geopolitical conditions.

Context captures the situation an item sits in, not its own activity. In the current approach, it comes from the item's structured attributes, its criticality, category, and supply class, the platform it supports, and that platform's lifecycle stage, and whether it is exercise sensitive, which are written into a short sentence and embedded alongside the other signals. For example, a demand rise on a critical, exercise-sensitive part on an aging platform means something very different from the same rise on a routine item, and context is what lets the model tell them apart. A richer context, such as regulatory or geopolitical conditions, would be added from real operational data.

4.2 Our Layered Analytical Architecture

Our approach consists of a layered analytical architecture, as shown in Figure 3 below.

What makes the architecture useful is the work that each layer does. The first layer profiles each entity separately, whether a supplier, an item, a depot, or a route. Most analytics stop at this level. In practice, a large share of supply chain and security risks do not surface when entities are examined one at a time. The second layer treats the relationship between two entities as something to model, since a dependency can weaken even when both entities remain healthy on their own. Two nominally independent suppliers that begin to move together are a relationship signal, not an entity signal. The third layer steps back to the whole network, where risk propagation, hidden common dependencies, and coordinated behavior become apparent only in the graph.

Taken together, the layers describe individual behavior, the links between entities, and the way risk spreads across the system.

Time is treated as a dimension that cuts across all three layers rather than as a fourth layer. Each entity, each relationship, and the network itself is stored as a series of time-stamped snapshots, and the model reads drift, velocity, and direction of change from that series. This is the part that separates the method from point-in-time reporting, because it lets the system act on where something is heading rather than only on its current state.

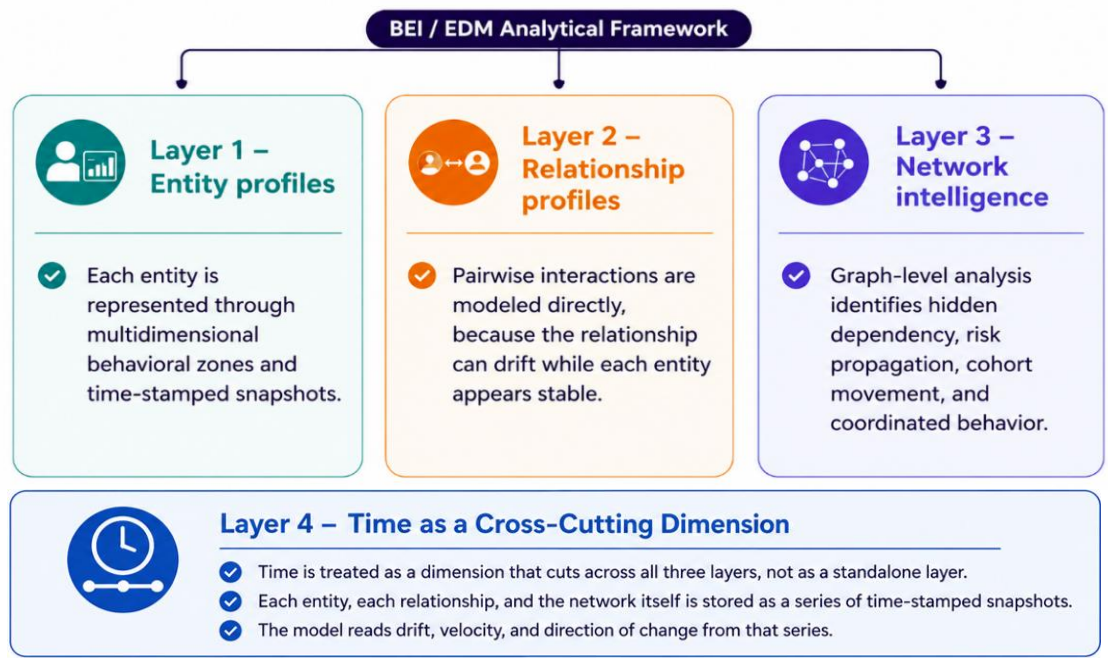


Figure 3- Our Layered Analytical Architecture

The architecture is best read as a direction of travel rather than a finished schema. Each layer is built to handle more data and attributes over time, including additional attributes at the entity layer, new relationship types at the relationship layer, and richer edges and propagation rules at the network layer. As the data foundation grows, the same three-layer structure can be widened and re-tuned without a redesign.

4.3 Mathematical and Model Design Principles

Positioning relative to existing tools: EDM / BEI is intended to complement, not replace, the established demand-planning and supply-chain-risk markets. Demand-sensing and planning platforms and supplier-risk and n-tier visibility platforms already provide control-tower visibility, probabilistic planning, and supplier monitoring. The differentiation claimed here is not any single technique but the composition: item-location behavioral drift, calibrated planning bands, peer-cohort transfer for sparse items, and a cross-domain early-warning method applied together at the application layer on DLA data. This positioning should be validated against incumbent tooling during the pilot.

Because every entity is represented as one consistent digital twin, the platform extends well beyond the three use cases shown here. The same representation supports many model families, from classification, survival, and anomaly models to graph, sequence, and trajectory methods, so increasingly advanced decision problems can be solved on one foundation. Beyond demand, supplier risk, and network early warning, the digital-twin layer applies to scenarios such as stockout and cascading-failure simulation, lead-time and allocation decisions, obsolescence and disposal timing, counterfeit screening, and readiness prioritization, with more advanced techniques, for example, trajectory models that anticipate where an entity is heading and scenario generation that stress-tests compound disruptions, added as the data supports them.

The three use cases in this paper are proof points; the digital-twin representation is the reusable foundation that makes this broader and more advanced capability possible.

5 Use Case 1- Demand Forecasting and Inventory Planning

Defense logistics demand is intermittent and highly variable, frequently driven by exercises, deployments, platform lifecycles, maintenance cycles, and supplier behavior. Deployments in particular concentrate demand around operational hot zones, the theaters and regions where activity surges and specific parts are consumed

far faster than baseline. The framework's space and context dimensions are built to read an entity's behavior relative to these zones, so a demand rise tied to a hot zone can be separated from ordinary variation. Richer deployment and theater data would sharpen this further.

As shown in Figure 4, the legacy demand approach remains important because it is transparent and fast. It examines one item at one depot, uses recent history, trend, exercise factors, and fiscal factors, then scales empirical quantiles through the same ratio. That makes it a valid baseline. Its structural limitation is the frozen band shape. Because P10 through P90 scale together, the P90 / P50 relationship does not adapt when demand calms, destabilizes, or shifts regime. Legacy also reacts backward, struggles with cold-start and sparse items, and cannot borrow behavior from similar items or related supply-chain entities.

Demand Forecasting and Inventory Planning

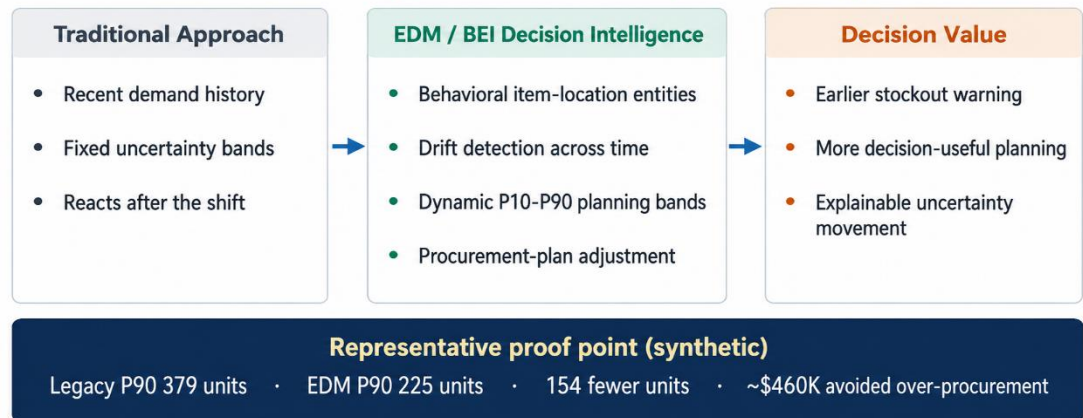
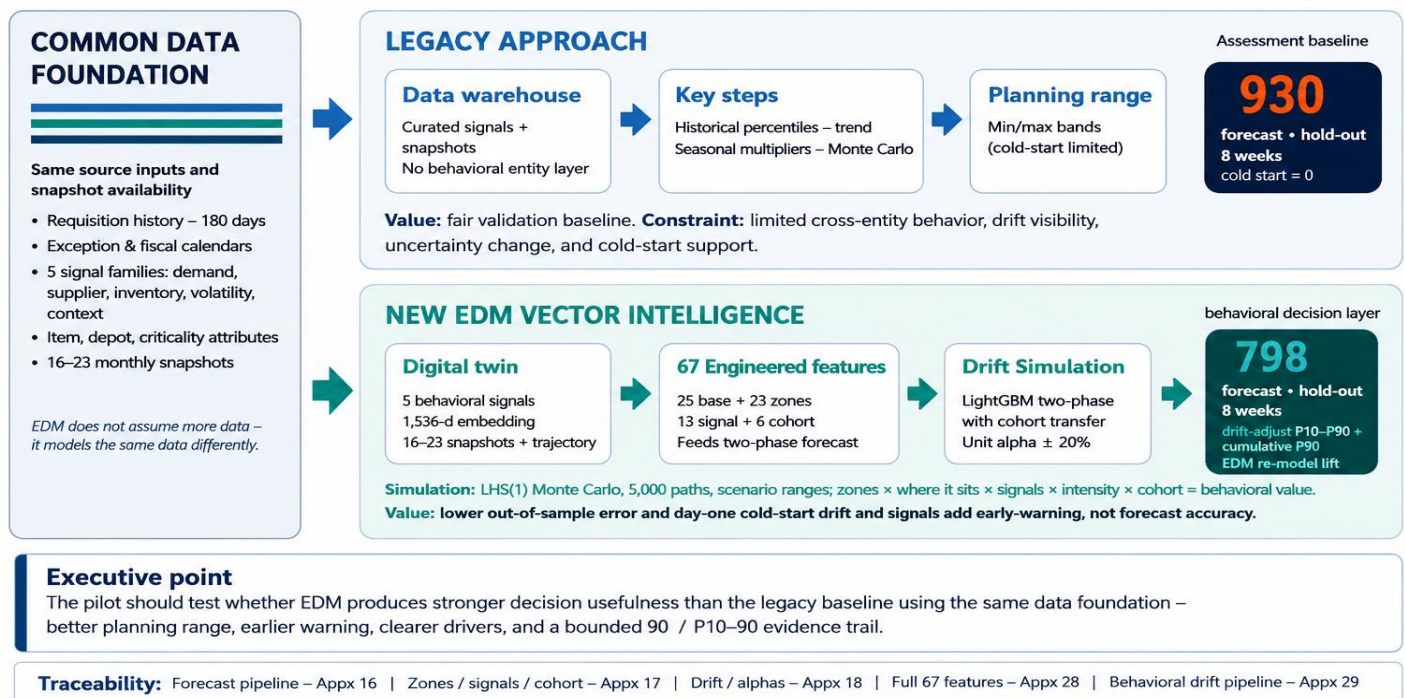


Figure 4: Demand Forecasting and Inventory Planning

EDM Vector Intelligence, shown in Figure 5, changes the unit of analysis. Each NSN x Department of Defense Activity Address Code (DODAAC) is modeled as a digital entity in a high-dimensional behavioral vector space.

Same data, stronger decision context

One shared data foundation feeds both paths. EDM re-models the same data as behavioral entities; the forecast lift comes from the embedding and base features, with cohort transfer for cold-start and drift, and signals for early warning, not forecast accuracy. Synthetic data



Executive point

The pilot should test whether EDM produces stronger decision usefulness than the legacy baseline using the same data foundation – better planning range, earlier warning, clearer drivers, and a bounded 90 / P10–90 evidence trail.

Traceability: Forecast pipeline – Appx 16 | Zones / signals / cohort – Appx 17 | Drift / alphas – Appx 18 | Full 67 features – Appx 28 | Behavioral drift pipeline – Appx 29

Figure 5: Legacy Approach and EDM Vector Intelligence

The entity is built from five behavioral signals: demand, supply, inventory, volatility, and context. Each month the

entity is re-embedded, producing 16 to 23 monthly snapshots that form a trajectory. The model then asks two questions: what has this item been doing, and where is it heading?

5.1 EDM Forecasting Pipeline

- **Digital twin:** The same records are re-modeled as a behavioral twin of each item and depot, as shown in Figure 6. Five signal families (demand, supply, inventory, volatility, and context) are serialized and embedded into 1,536-dimensional

Inside the new approach: one digital twin → 67 engineered features

The same records become a behavioral twin of each item-sku, distilled into the features the two-phase model learns from — then nudged for where it's heading.

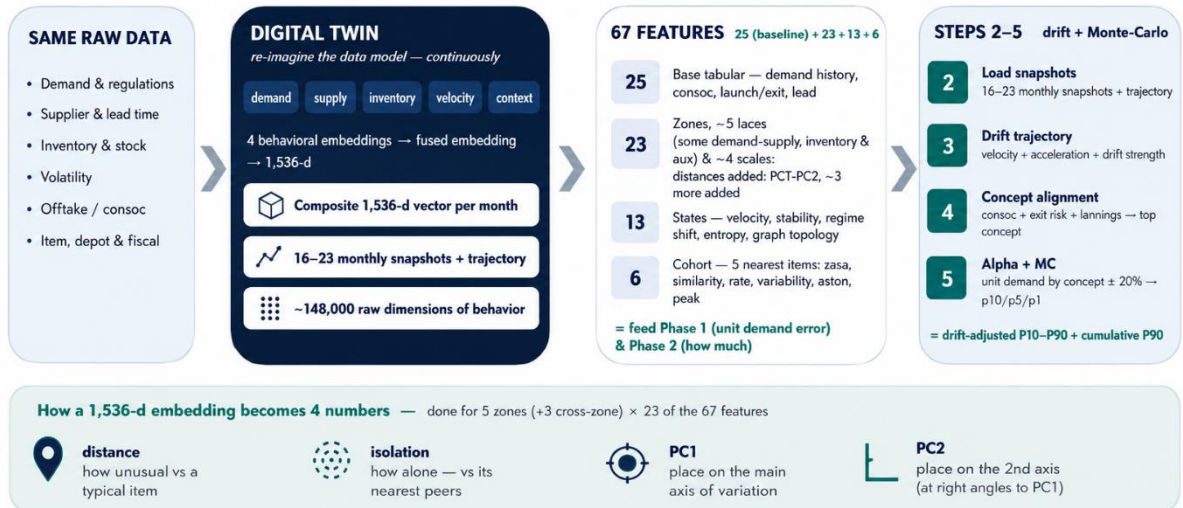


Figure 6: Digital Twin Demonstration

- vectors, composited into one monthly vector, with 16 to 23 monthly snapshots forming the trajectory.
- **Engineered features:** The twin is distilled into up to 67 features: 25 base tabular (demand history, exercise, item and depot, fiscal), 23 zone-derived (each behavioral zone reduced to its distance, isolation, and first two principal components, plus cross-zone terms), 13 drift signals, and 6 cohort features drawn from the nearest behavioral twins. The measured configuration used 29 of these features; the full 67-feature architecture activates as more data becomes available. Of the engineered features, 42 (the zone, drift, and cohort families) are derived from the embedding rather than from raw tabular fields, which is the embedding's distinct contribution; a tabular-only variant (V1, 25 features) and the embedding-enriched variant (V2, 29 features) are compared to test that contribution directly.
- **Cohort transfer:** For items with little history, a nearest-behavioral-twins (k-NN) transfer produces a usable day-one forecast, addressing cold-start.
- **Baseline:** A two-stage LightGBM hurdle model separates occurrence from quantity. The first stage estimates whether demand occurs. The second stage estimates P10 through P90 quantiles and enforces monotonic quantile order.
- **Snapshots:** 16 to 23 monthly composite snapshots are loaded for each entity. If fewer than two exist, the system returns the baseline unchanged, which provides graceful degradation.
- **Drift trajectory:** Euclidean (L2) distances between consecutive snapshots generate velocity, acceleration, mean-step drift, consistency, drift intensity, and drift norm.
- **Concept alignment:** Drift direction is projected against eight pre-embedded supply-chain concepts by cosine similarity, so the model can name what the entity is becoming.
- **Alpha and dynamic bands:** The model converts concept alignment into a bounded plan adjustment and a band multiplier, then reruns Monte Carlo to produce adjusted quantiles and 8-week safety-stock targets.

5.2 Demand Planning Decision-Value Evidence

The current proof-of-concept results show that EDM translates the same demand, supply, inventory, volatility, and context signals into calibrated planning ranges and right-sized procurement, reducing over-procurement on items the baseline over-orders and catching demand the baseline reports as zero, as quantified above. These are decision-value results on synthetic data, not an operational claim. The next step is to validate them on DLA

operational data through a bounded pilot that compares EDM / BEI outputs against existing baselines and SME judgment.

The worked examples below show both failure directions on representative items (8-week P90 forecasts). These worked-example forecasts and the calibration figures reported below come from the validated embedding-enriched configuration (V2, 29 features); the 67-feature count is the full architecture, of which not all layers were active in these measurements.

Item (NSN)	Legacy 8-wk P90	EDM 8-wk P90	Outcome
Adapter plate (5120-01-397-8469)	379	225	40% smaller buy, about \$460K less over-procurement
Needle bearing (3110-01-614-2738)	1,184	545	54% smaller buy, about \$48K less over-procurement
Valve train kit (2910-01-905-6139)	0	95	catches demand legacy reports as zero, about \$28K

Measured against realized demand on these items, EDM's planning bands track actual requisitions far more closely than the legacy bands. For the adapter plate, the EDM 8-week P90 of 225 units sits at the observed peak 8-week demand of 224 units, while the legacy P90 of 379 carried roughly 155 units of excess. For the needle bearing, the EDM P90 of 545 is far nearer realized demand than the legacy P90 of 1,184, which over-forecasts by several hundred units. In each case, EDM right-sizes toward real demand rather than padding the band.

A separate held-out test measures forecast accuracy directly: the last eight weeks of demand were hidden, each method forecasted them blind, and the forecasts were compared to actual demand. On a representative tracked-vehicle part, EDM's eight-week forecast of 798 units was closest to the actual 627 units, against 930 for the moving-average baseline and 903 for a time-series foundation model, a held-out Weighted Absolute Percentage Error (WAPE) of 0.78 for EDM versus 0.95 and 0.91. Across the other items tested, EDM held the lower held-out error against the moving-average baseline, for example ballistic panel 0.33 versus 0.69, sleeve bearing 0.44 versus 0.57, protective insert 0.76 versus 0.82, and armor mounting bracket 0.63 versus 0.66. These remain synthetic-data results to be re-validated on DLA data.

Because the proof-of-concept data carries an item criticality code (the adapter plate and needle bearing are coded C, and the valve train kit B), the planning quantile can be tiered by criticality: a higher service level (for example, P95 to P99) for sole-source, critical, or war-reserve items, and a lower level for routine items, rather than a uniform P90.

Note: *The National Stock Numbers shown in these examples are illustrative synthetic identifiers used for the proof of concept; they do not correspond to actual catalog records.*

Technical Point	Source Detail	Decision Interpretation
Synthetic scope	500 items, 200 suppliers, 4 depots, 24 months, approximately 277K requisitions	Large enough for concept testing, but still requires DLA-data remeasurement
Entity representation	high-dimensional vectors (current pilot: 1,536-d), five signals, 16-23 monthly composites	Forecasting becomes trajectory-aware rather than row-based
Feature strategy	25 base tabular features (V1); 29 in the measured embedding-enriched variant (V2); up to 67 with all EDM layers active	Technical specificity without overstating measured configuration
Calibration	Occurrence ECE 0.0322 and Brier score 0.0885, Grade A	Probability outputs are credible enough to test safety-stock planning
Quantile coverage	P10 0.141, P25 0.257, P50 0.498, P75 0.743, P90 0.894	Quantile coverage is close to nominal, with conservative lower-band behavior
Procurement example	Adapter Plate NSN 5120-01-397-8469 at W25H1R, legacy P90 379 vs. EDM P90 225	154 fewer P90 units, approximately \$460,000 avoided over-procurement on one item

Disruption example	Sole-source critical item aligns with supply_disruption, alpha capped at +30%, bands widen 1.4x	Potential 1-3 month head start before disruption appears in delivery counts
Volatility example	Volatility_increase concept widens bands 1.6x with median unchanged	Model signals uncertainty without pretending to know direction
Operational safeguards	Alpha capped at +/-30%, weak alignment zeroed, inconsistent drift damped, exercise-sensitive rising demand protected	Prevents unstable procurement movement from noisy drift

The demand use case should be evaluated on procurement-plan accuracy, calibration, pinball loss, mean absolute scaled error (MASE), horizon-level performance, early warning, and planner usefulness. Point-accuracy metrics should remain visible, but they should not become the only scorecard because the EDM value proposition is the distribution and its direction, which is a more useful basis for planning.

6 Use Case 2 - Supplier Risk and Supply Chain Risk Management (SCRM)

We understand Supply Chain Risk Management (SCRM) as a compliance-driven, mission assurance discipline that must operate across the full supplier, technology, logistics, and service lifecycle. Our approach aligns SCRM governance with federal acquisition, cybersecurity, and supply-chain assurance requirements, including NIST SP 800-161 and related federal requirements. It emphasizes supplier qualification, component authenticity, anti-counterfeit inspection, and cybersecurity validation, with continuous updates to the SCRM plan as supplier profiles, regulations, threat intelligence, or operational conditions change.

Supplier Risk and Supply Chain Risk Management



Figure 7: Supplier Risk and Supply Chain Risk Management

For DLA, this compliance foundation is directly relevant to supplier risk, mission assurance, and supply-chain risk mitigation in a contested logistics environment. Adversarial exploitation and disruption may emerge through counterfeit or prohibited components, hidden sub-tier exposure, sourcing shifts, cyber-enabled compromise, supplier concentration, delivery disruption, or regulatory noncompliance before conventional supplier metrics show failure. Our SCRM posture provides the governance and control foundation for identifying, assessing, mitigating, and reporting these risks. EDM / BEI builds on that foundation by modeling suppliers, NSNs, locations, routes, manufacturers, contracts, countries of origin, and sub-tier relationships as connected behavioral entities, enabling DLA to evaluate where supplier behavior and supply-chain exposure are moving before readiness or operational resiliency is affected. Supplier risk, as shown in Figure 7, is better treated as a continuous mission and readiness consideration than as a periodic compliance check, because it can develop while current-period supplier metrics remain acceptable. A supplier may continue to deliver within threshold, pass quality checks, and meet price expectations while changing its sourcing origins, sub-tier dependencies, manufacturer relationships, or concentration exposure. Conventional monitoring confirms observed failure and frequently does not detect such drift.

BEI for supplier risk applies the same entity principle to suppliers, NSNs, locations, routes, manufacturers, countries of origin, contracts, financial indicators, and sub-tier relationships. It asks what the supplier is becoming, how its relationships are changing, and whether those changes are unusual for its cohort. That matters because many supplier risks appear in relationships before they appear in aggregate supplier metrics.

6.1 Supplier Behavioral Zones and Relationships

Our Supplier Risk Zones and Relationships are detailed in Figure 8, below. This is the three-layer Behavioral Entity Intelligence architecture applied to supplier risk. Rather than reduce a supplier to a single score, EDM Vector Intelligence represents each supplier as a behavioral entity described across several zones, and it treats the connections between suppliers, parts, and locations as objects in their own right. Figure 8 shows both. The top row is the entity layer: five behavioral zones, each tracking a different aspect of the supplier (identity, performance trajectory, geographic and sourcing exposure, network position, and financial standing) and the risk that surfaces when that zone drifts. The lower half is the relationship and network layer: it maps how suppliers, NSNs, countries of origin, depots, and bills of materials connect, where exposures such as hidden sub-tier dependency or adversary-nation sourcing become visible only when entities are viewed together.

Supplier Risk Zones and Relationship Map

Behavioral zones and linked relationships that expose supplier risk

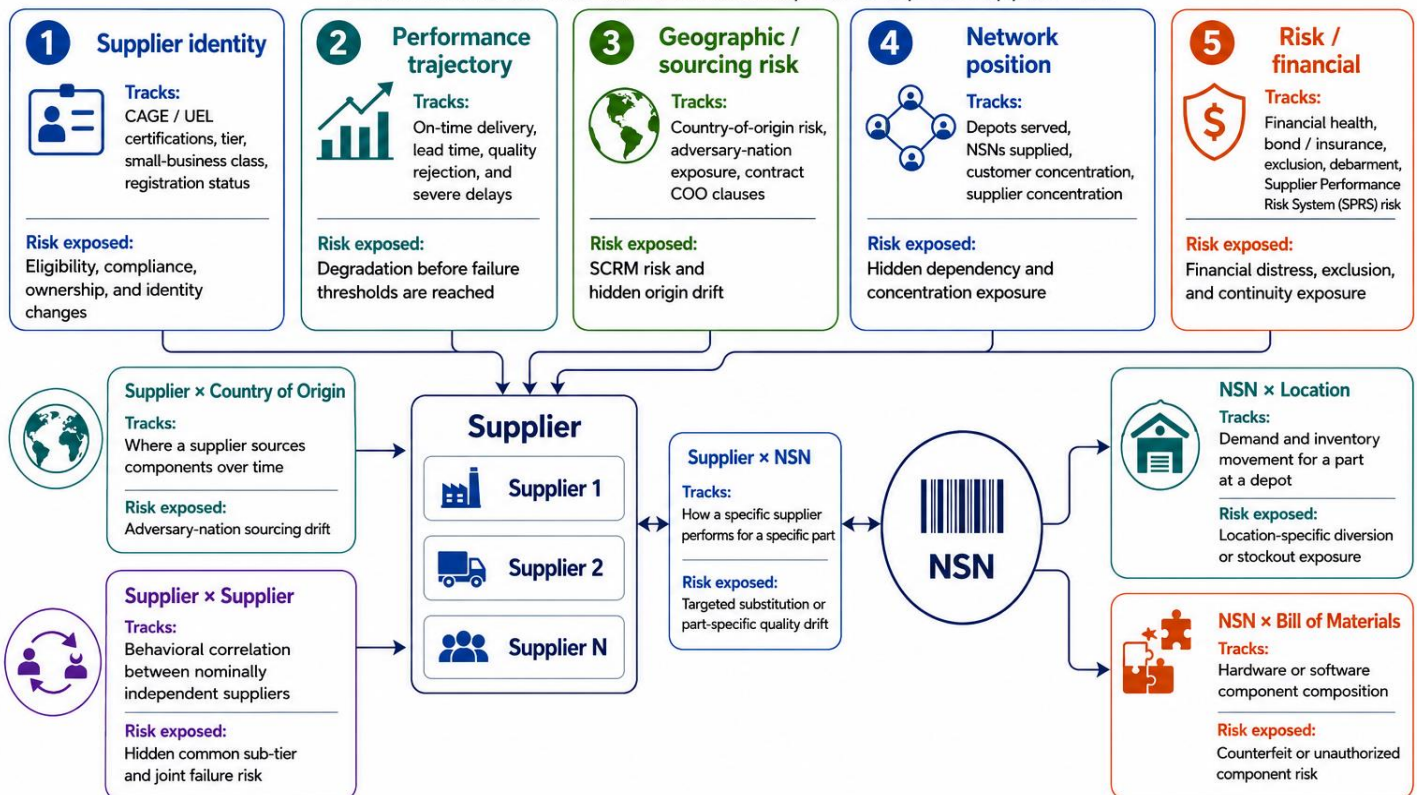


Figure 8: Our Supplier Risk Zones and Relationship Map (Target Architecture)

6.2 Supplier-Risk Technical Proof Points

In a proof-of-concept modeling phase, a supplier-risk classifier and a survival model were trained on synthetic data. The early results show that the behavioral approach transfers from demand to supplier risk: the models flag higher-risk suppliers and estimate time-to-failure with explainable drivers. Applied as a full behavioral model, the same digital-twin representation opens up much more, including relationship and sub-tier exposure, cohort movement, and drift that surfaces before conventional supplier metrics react, giving DLA earlier and more connected supplier-risk insight.

6.3 Supplier Risk Detection Scenarios

Scenario	Traditional Visibility	BEI Signal
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Supplier sourcing shift to adversary nation	May remain invisible until annual audit or shipment review	The geographic risk zone drifts while the performance zone stays stable
Hidden supplier concentration	Suppliers appear independent when reviewed separately	Network cohort analysis flags correlated supplier behavior
NSN transitioning to stockout	Visible only when a stockout occurs	Inventory position zone accelerates toward depletion regime
Counterfeit part entry	Often detected after a field failure or inspection issue	Supplier x NSN failure divergence and component relationship drift
Supplier financial distress	Detected at the next quarterly or external review	Performance and financial zones drift concurrently

The supplier use case should be validated as a risk-intelligence capability, not merely as a replacement for existing supplier scores. Current systems remain important. BEI adds continuous behavioral direction, relationship-level visibility, calibrated failure probability, timing through survival modeling, and explanations that can be reviewed before risk becomes delivery failure or readiness impact.

7 Use Case 3 - Logistics Network Early Warning and Mission Readiness Risk

DLA manages an interconnected network of NSNs, depots, suppliers, routes, and platforms in which readiness risk can build quietly across multiple entities simultaneously. A part's availability can erode through a chain of small shifts, a supplier slipping, a depot buffer thinning, a route congesting, or a shared sub-tier wobbling, none of which crosses a threshold on its own, until the result appears as a stockout or a mission-readiness gap. This is the risk Use Case 3 targets: slow, directional, relationship-level change that no single system surfaces.

The mechanism for this already exists in the platform. The graph-based risk-propagation and cross-entity correlation components operate on logistics entities, linking NSNs, depots, suppliers, routes, and sub-tier dependencies into a single network and tracing how risk flows through it. What remains to be quantified in the DLA data is how early and reliably the network view warns of readiness risk.

The underlying detection principle, finding slow, hidden, directional change that stays within conventional thresholds, has been validated most rigorously in a cybersecurity proof of concept. There, the same behavioral approach detected all four long-duration intrusion campaigns (4 of 4) at an 8.1 percent false-positive operating point, while traditional anomaly detection and signature-based tools caught zero to one. The two domains differ in data, but they share the structure that the framework is built to detect, which is why that result is meaningful evidence that the same mechanism can surface slow, networked, sub-threshold risk in logistics. This is offered

Logistics Network Early Warning and Mission Readiness Risk

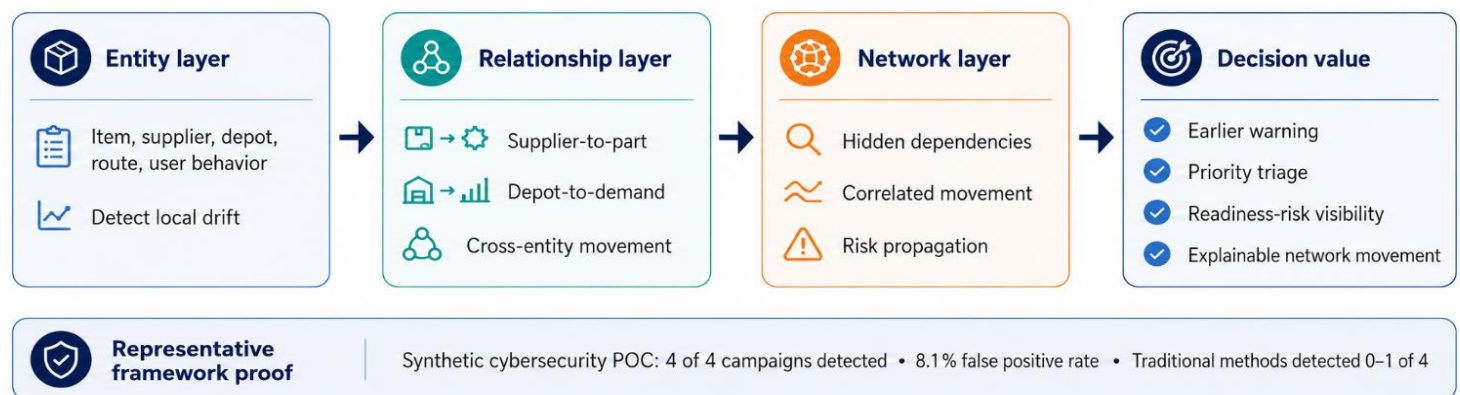


Figure 9: Logistics Network Early Warning and Mission Readiness Risk

as a transferable detection principle, that slow, persistent, sub-threshold signals are discoverable, and as a hypothesis to validate on DLA logistics data, not as a logistics performance result.

This is where the three-layer architecture shown in Figure 9 earns its place. Entity profiles flag drift in a single item, supplier, or depot. Relationship profiles catch pairwise movement that a single-entity review misses, such as two suppliers drifting together. Network intelligence surfaces what only the graph shows: hidden dependencies, correlated movement, risk propagation, and coordinated behavior. For DLA, this is the basis of an early-warning lane for replacement-part readiness, hidden supplier concentration, depot-specific stockout exposure, and route-disruption sensitivity, with software bill of materials (SBOM) drift and broader mission-facing logistics risk added as the framework extends to richer data. How early and how reliably it warns on each of these is what the validation pilot would measure.

7.1 Approach Proof and Transfer Logic

Proof Area	Source Detail	DLA Transfer Logic
Cyber proof of concept population	250 users across employees, contractors, and service accounts	Demonstrates entity profiling at the population scale
Observation period	485 days from January 2025 to May 2026	Shows long-duration temporal drift tracking
Event volume	Approximately 14 million events across 5 log sources	Demonstrates capacity to process large behavioral telemetry
Attack campaigns	4 long-duration campaigns: insider threat, slow APT command-and-control, living-off-the-land intrusion, telecom-sector espionage	Shows ability to detect slow, sophisticated movement
Detection result	4 of 4 campaigns detected at 8.1% false positive rate	Shows directional scoring can detect hidden behavior
Traditional comparison	Z-score detected 1 of 4, LOF / Isolation Forest / OCSVM detected none	Shows magnitude-only methods missed directional movement
Triage value	Zone-level directional intelligence identified which behavioral dimension changed	Shows outputs can be explained, not just scored
Prior-art discipline	Novelty is composition: semantic representation, direction, cohort, relationship, network, and multi-phase scoring	Protects claim credibility and avoid overstating any single method

Note: These cybersecurity results establish the detection principle by surfacing slow, directional change that remains within conventional thresholds. The same behavioral signals underpin the logistics use cases, so the principle carries over; the specific detection and false-positive rates would be measured on DLA data.

7.2 DLA Mission Patterns Supported by BEI

- **Replacement-part readiness exposure:** Part demand, supplier capacity, inventory position, and platform context drift toward risk before a visible shortage.
- **Hidden common dependency:** Nominally independent suppliers show correlated behavior because they share a sub-tier source, geography, process, or manufacturer.
- **Route or location sensitivity:** NSN x depot or supplier x location behavior changes in ways that indicate theater, route, or depot-specific exposure.
- **SBOM or component drift:** Bill-of-materials composition changes without clear notification, creating counterfeit, compliance, or security risk.
- **Demand diversion or fraud pattern evolution:** Demand composition, recipient location, or item relationship changes while aggregate demand volume appears normal.
- **Cohort movement:** Multiple entities in the same peer group drift together, pointing to shared causes such as market shock, geopolitical exposure, or supplier ecosystem stress.

This use case should not be positioned as a broad “detect everything” capability. It should be validated as an early-warning and triage layer for selected mission scenarios where existing DLA data can populate entities, relationships, cohorts, and networks. The strongest initial candidates are

scenarios where individual metrics have historically remained acceptable until the risk was already operationally consequential.

8 Technical Evidence and Validation Boundaries

This section distinguishes proof of concept from operational proof. The current proof of concept is strong because it does more than describe a concept. It demonstrates that EDM / BEI can model DLA-relevant demand, supplier-risk, and network early-warning scenarios as shown in Figure 10 using controlled synthetic data, measurable baselines, calibrated outputs, explainable drivers, and defined validation boundaries. That level of evidence is sufficient to warrant a bounded pilot. The remaining question is not whether the concept is technically plausible. The remaining question is whether the same approach produces measurable mission value on DLA operational data.

The pilot should therefore be treated as the logical next validation step. It should test EDM / BEI in advisory, side-by-side mode against current baselines, SME review, and agreed success measures. The objective is to determine whether EDM /

Technical Evidence Summary

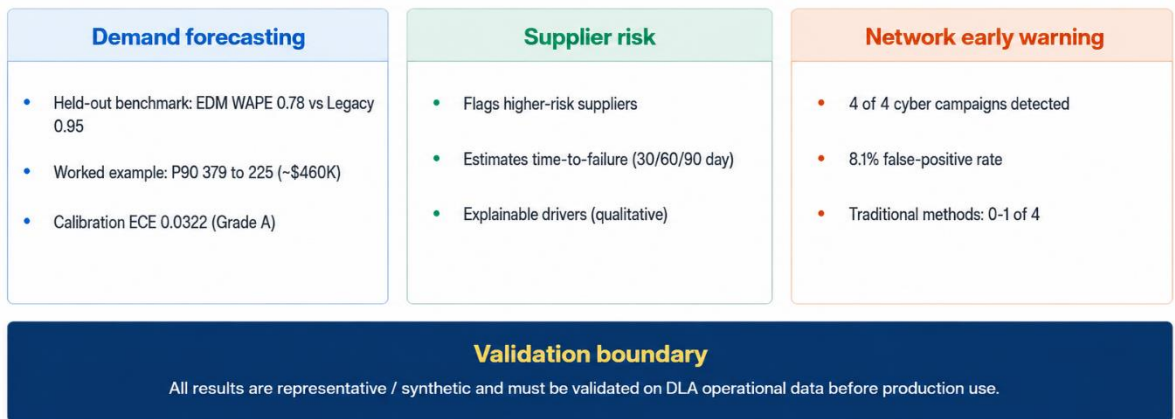


Figure 10: Technical Evidence Summary

BEI improves decision usefulness in DLA's operating environment, including forecast calibration, procurement-plan accuracy, early-warning lead time, supplier-risk discrimination, relationship-level visibility, explainability, analyst usability, and integration with existing workflows. This framing protects claim credibility while giving DLA a clear, risk-managed basis for moving from proof-of-concept evidence to operational data validation.

Evidence Category	What the Evidence Shows	How It Should Be Claimed
Demand forecasting	EDM creates calibrated, dynamic uncertainty bands and can materially alter procurement planning in representative synthetic scenarios	Claim as a strong basis for DLA data validation, not as operational superiority
Procurement value	Adapter Plate example shows 154 fewer P90 units and approximately \$460K avoided over-procurement	Use as a representative synthetic proof of decision-value potential
Supplier risk	Behavioral modeling flags higher-risk suppliers, estimates time-to-failure, and explains its drivers	Claim as a qualitative proof of concept requiring real-data validation
Network early warning	BEI detected all 4 embedded cyber campaigns at 8.1% FP while traditional methods caught at most one	Claim as framework evidence and transfer logic, not a DLA logistics production result
Architecture	high-dimensional embeddings, cohort baselining, concept projection, relationship modeling, graph intelligence, and multi-phase scoring create a distinct composition	Claim composition-level differentiation, not novelty of every component
Deployment readiness	Containerized FastAPI, PostgreSQL / pgvector, HNSW index, SCD2 history, gold views, Docker, and air-gap option	Claim practical implementation ability for controlled validation



Data dependency	Existing DLA requisition, inventory, procurement, supplier, quality, COO, SBOM, and financial feeds can populate the model	Validate data access, quality, entity resolution, and governance before pilot expansion
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8.1 Technical Evidence Summary

- EDM demand forecasting uses a two-stage hurdle LightGBM model with occurrence classification and quantile regressors for P10 through P90.
- Demand entities are represented in a high-dimensional semantic vector space using demand, supply, inventory, volatility, and context signals.
- Demand Envelope construction captures identity, time series, demand decomposition, supply network, inventory position, financial profile, operational context, and embeddings.
- EDM drift uses velocity, acceleration, mean-step drift, consistency, concept alignment, bounded alpha, and dynamic band multipliers of approximately 0.85x to 1.6x.
- Demand safeguards include +/-30% alpha cap, weak-alignment zeroing, inconsistency dampening, and protection for exercise-sensitive rising demand.
- Supplier risk uses entity profiling, zone decomposition, relationship analysis, temporal trajectory, regime detection, XGBoost classification, Random Survival Forest timing, SHAP explanation, and monotone constraints.
- BEI framework uses nine dimensions, three analytical layers, self-baseline and cohort-baseline comparisons, concept naming, CUSUM-style slow-drift accumulation, relationship profiles, and multi-phase fusion.
- The cybersecurity proof of concept remains important because it demonstrates the framework's ability to detect directional movement that traditional magnitude-based methods missed.

9 Recommended Pilot Validation Path

The next step should be advisory, side-by-side validation as shown in Figure 11 on DLA operational data. The purpose is not to displace current systems in one move. The purpose is to determine whether EDM/BEI provides measurable decision value beyond current baselines for selected logistics decisions. That path protects DLA's mission, preserves the discipline, and creates an evidence-based basis for pilot expansion if results warrant it.

Bounded Pilot Validation Path

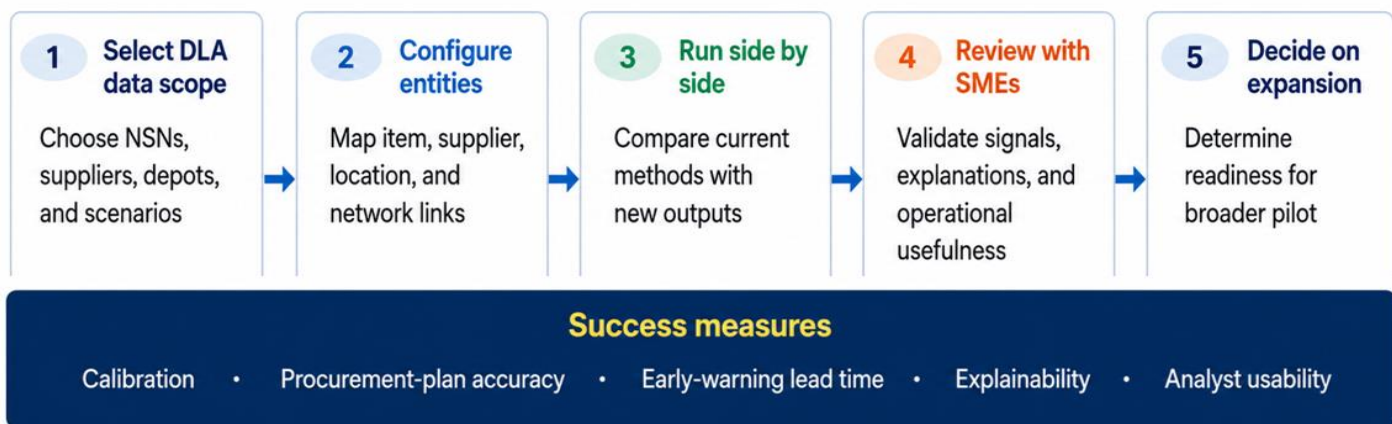


Figure 11: Bounded Pilot Validation Path

The pilot should run in parallel with existing planning and monitoring processes. It should not feed automated procurement or supplier action until results are validated by subject matter experts (SMEs) and decision owners. The initial scope should be narrow enough to execute quickly and broad enough to test the three use cases, in priority order, meaningfully.

Pilot Element	Recommended Scope	Success Measures
Demand forecasting	Selected NSNs, depots, high-value intermittent items, exercise-sensitive items, and sole-source critical items	Calibration, pinball loss, MASE, procurement-plan accuracy, dynamic band usefulness, and early warning

Supplier risk / SCRM	Selected suppliers, commodity groups, NSNs, COO / SBOM records, quality, and delivery history	Risk discrimination, survival timing, explainability quality, relationship coverage, false-positive review, SME usability
Network early warning	Supplier x NSN, NSN x Location, Supplier x Country, Supplier x Supplier, BOM relationships	Relationship drift detection, hidden dependency identification, risk propagation, triage usefulness
Data readiness	Requisitions, orders, inventory, procurement, delivery, quality, SPRS, Product Data Reporting and Evaluation Program (PDREP), Government-Industry Data Exchange Program (GIDEP), SAM.gov, PIEE / WAWF, DIBBS, COO, SBOM, financial feeds	Completeness, data lineage, entity resolution, timeliness, access control, security posture
Governance	SME review board, model audit, claim discipline, threshold tuning, operational review cadence	Confidence in outputs, explainability, operational acceptance, and bounded decision use
Pilot expansion decision	Evidence reviews after side-by-side validation window	Proceed, refine, expand, or stop based on measured mission value

9.1 Suggested Pilot Operating Model

- Operate in read-only advisory mode during the initial validation window.
- Run existing baselines and EDM / BEI outputs side by side so DLA can compare signal quality without disrupting operations.
- Use historical back-testing and forward-looking shadow mode to evaluate both retrospective and live decision usefulness.
- Require SME review of explanations before treating a signal as operationally meaningful.
- Publish validation results with a clear distinction between baseline performance, EDM / BEI performance, synthetic carryover assumptions, and real-data findings.
- Maintain model and data governance from the start, including provenance, security, access control, calibration monitoring, and documentation of limitations.

10 Implementation Considerations

The implementation details for EDM / BEI will be addressed in a follow-on implementation strategy paper. This innovation concept paper establishes the mission need, technical proof-of-concept, validation boundaries, and a recommended pilot path. The next paper will translate that foundation into an executable implementation plan that DLA can use to evaluate scope, schedule, data requirements, security posture, governance, success measures, level of effort, and the appropriate acquisition pathway. It will also name the target integration surface, including the Enterprise Business System (EBS), the demand-planning system, the DLMS transaction layer, and the closed-loop path by which a validated planning band re-enters the system of record rather than remaining a side-by-side view.

The follow-on paper will define the structure and execution of the recommended pilot, including the initial use-case scope, data-access assumptions, system integration approach, deployment options, security and authorization considerations, model governance, SME review process, validation scorecard, delivery timeline, and transition criteria. It will also support acquisition planning by outlining how it could be pursued through an appropriate task order, Commercial Solutions Opening (CSO), or other contractual vehicle selected by DLA and its contracting office.

11 Conclusion

DLA's highest-value logistics decisions increasingly depend on detecting movement before failure is visible. In a contested logistics environment, adversarial exploitation and disruption can affect demand, supplier behavior, sourcing exposure, route vulnerability, sub-tier dependency, cyber-enabled continuity, inventory posture, and correlated network risk before conventional thresholds are breached. For DLA, adapting faster, resolving supply-



chain friction earlier, and preserving decision time are directly tied to warfighter readiness, mission assurance, operational resiliency, and U.S. military superiority. This is the decision environment EDM / BEI is designed to address.

Based on the assessment in this paper, we recommend that DLA begin with Use Case 1 - Demand Forecasting and Inventory Planning as the initial pilot focus. This use case provides the clearest path to validate EDM / BEI against existing demand, supply, inventory, and procurement-planning data while producing measurable outputs tied to forecast quality, uncertainty management, procurement-plan accuracy, early warning, and decision advantage. The pilot scope, assumptions, data-access requirements, schedule, level of effort, and success measures should be refined jointly with DLA and confirmed through mutual agreement before execution.

Validated against DLA operational data, EDM/BEI is not another model or dashboard. It is a decision-intelligence layer that reveals where demand, supplier behavior, and network risk are heading before the consequence is visible, giving DLA the decision advantage that defines readiness in a contested logistics environment: act earlier, buy smarter, and decide faster.